Logistic Regression + G.D.

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rows = m columns = n. (m,n).

					y	$\hat{y}$
x_{11}	x_{12}	x_{13}	------	x_{1n}	y_1	$\hat{y}_1$
x_{21}	x_{22}	x_{23}	----	x_{2n}	y_2	$\hat{y}_2$
⋮						
x_{m1}	x_{m2}	x_{m3}	------	x_{mn}	y_m	$\hat{y}_m$

$$\hat{y}_i = \sigma(z) \rightarrow z = \sum w_i x_i$$

$$\hat{y}_1 = \sigma(w_1 x_{11} + w_2 x_{12} + \text{-----} + w_n x_{1n} + w_0)$$
$$\hat{y}_2 = \sigma(w_1 x_{21} + w_2 x_{22} + \text{------} + w_n x_{2n} + w_0)$$

$$\hat{Y} = \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_m \end{bmatrix} = \begin{bmatrix} \sigma(w_0 + w_1 x_{11} + w_2 x_{12} + \text{----} + w_n x_{1n}) \\ \vdots \\ \sigma(w_0 + w_1 x_{m1} + w_2 x_{m2} + \text{----} + w_n x_{mn}) \end{bmatrix}$$

$$\hat{Y} = \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_m \end{bmatrix} = \sigma\left(\begin{bmatrix} w_0 + w_1 x_{11} & \text{-------} \\ \vdots & \vdots \\ w_0 + w_1 x_{m1} + & \text{---} + w_n x_{mn} \end{bmatrix} \right)$$

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$$\hat{y} = \sigma\left(\begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1n} \\ 1 & x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & \vdots & & & \vdots \\ 1 & x_{m1} & x_{m2} & \cdots & x_{mn} \end{bmatrix} \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_n \end{bmatrix}\right)$$

$$\hat{y} = \sigma(X.W) \quad \text{—— (i)}$$

we have : loss f^n:

$$L = \frac{-1}{m} \sum_{i=1}^{m} \left(y_i \log(\hat{y}_i) + (1-y_i) \log(1-\hat{y}_i) \right)$$

or, $$L = \frac{-1}{m} \left[\sum_{i=1}^{m} y_i \log(\hat{y}_i) + \sum_{i=1}^{m} (1-y_i) \log(1-\hat{y}_i) \right] \quad \text{—— (ii)}$$

$$\sum_{i=1}^{m} y_i \log(\hat{y}_i) = y_1 \log(\hat{y}_1) + y_2 \log(\hat{y}_2) + \cdots + y_m \log(\hat{y}_m)$$

$$= [y_1 \; y_2 \cdots y_m] \begin{bmatrix} \log(\hat{y}_1) \\ \log(\hat{y}_2) \\ \vdots \\ \log(\hat{y}_m) \end{bmatrix}$$

$$= Y \cdot \log \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_m \end{bmatrix}$$

$$= Y \cdot \log \hat{Y}$$

$$= Y \cdot \log(\sigma(X.W)) \quad \text{—— (iii)}$$

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Ally, $\sum_{i=1}^{m} (1-y_i) \log(1-\hat{y}_i) = (1-Y) \cdot \log(1-\hat{Y})$ — (iv)

Now; (ii) becomes: [(iii) & (iv)]

$$\boxed{\begin{aligned} \text{Loss} &= \frac{-1}{m}\left[Y \log \hat{Y} + (1-Y) \log(1-\hat{Y}) \right] \\ &\text{where } \hat{Y} = \sigma(X \cdot W) \quad \text{— [i]} \end{aligned}}$$

↳ reqd loss f^n in matrix term.

for i in epoch:
$$w = w - \eta \cdot \frac{\partial L}{\partial w}$$
} Gradient descent.

$$L = \frac{-1}{m}\left[Y \log \hat{Y} + (1-Y) \log(1-\hat{Y}) \right]$$

$$= \cancel{\frac{-1}{m}\left[Y \log \right.}$$

$$\frac{\partial L}{\partial w} = \frac{-1}{m} \cdot \frac{\partial}{\partial w}\left[Y \log \hat{Y} + (1-Y) \log(1-\hat{Y}) \right].$$

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$$\frac{\partial (Y \log \hat{Y})}{\partial w} = \frac{Y}{\hat{Y}} \cdot \frac{\partial \hat{Y}}{\partial w}$$

$$= \frac{Y}{\hat{Y}} \cdot \frac{\partial (\sigma(X.w))}{\partial w}$$

$$= \frac{Y}{\hat{Y}} \cdot \left[\sigma(X.w)(1-\sigma(X.w)) \right] \cdot X$$

$$= \frac{XY}{\hat{Y}} \cdot \hat{Y}(1-\hat{Y})$$

$$= XY(1-\hat{Y})$$

$$\frac{\partial [(1-Y)\log(1-\hat{Y})]}{\partial w} = \frac{(1-\hat{Y})}{1-\hat{Y}} \cdot (-1) \cdot \frac{\partial \hat{Y}}{\partial w}$$

$$= \frac{1-Y}{1-\hat{Y}} \cdot (-1) \cdot \frac{\partial [\sigma(X.w)(1-\sigma(X.w))] \cdot X}{\partial w}$$

$$= -\frac{(1-Y)}{\cancel{1-\hat{Y}}} X \cdot \hat{Y} \cdot \cancel{(1-\hat{Y})}$$

$$= -X\hat{Y}(1-Y)$$

$$\frac{\partial L}{\partial w} = \frac{-1}{m} \left[XY(1-\hat{Y}) - X\hat{Y}(1-Y) \right]$$

$$= \frac{-1}{m} \left[Y(1-\hat{Y}) - \hat{Y}(1-Y) \right] X$$

$$= \frac{-1}{m} \left[Y - \cancel{Y\hat{Y}} - \hat{Y} + \cancel{\hat{Y}Y} \right] X.$$

$$\boxed{\frac{\partial L}{\partial w} = -\frac{1}{m}(Y-\hat{Y})X}$$

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Now;

$$w = w - \eta \cdot \frac{\partial L}{\partial w}$$

$$\frac{-1}{m}(Y - \hat{Y}) \cdot X$$

$$\therefore \; w = w + \eta \cdot \frac{1}{m}(Y - \hat{Y}) \cdot X$$

$$w = \begin{bmatrix} w_0 \\ w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix}_{(n+1)\times 1} \quad X = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1n} \\ 1 & & & & \vdots \\ \vdots & & & & \vdots \\ 1 & \cdots & \cdots & \cdots & x_{mn} \end{bmatrix}_{m\times(n+1)} \quad Y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix}$$

$$\hat{Y} = \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_m \end{bmatrix}$$

Code for from stiach:

$$w = w + \eta \cdot \frac{1}{m}(Y - \hat{Y}) \cdot X$$

$$\hat{Y} = \sigma(X \cdot w) \qquad \sigma = \frac{1}{1 + e^{-z}}$$

$$z = X \cdot w$$